Protocol B results: thresholds selected on validation

This report gives the Protocol B evaluation of the trained HybridSBD ablation models (leak-free SHOT training set of 448 videos). In Protocol B the decision thresholds are **not** tuned on the test sets: for each model, τ_c and τ_g are selected once, by maximising F1 on the 67 SHOT validation videos, and then applied unchanged to the SHOT test set (200 videos), the ClipShots test set (500 videos) and BBC (11 episodes). The minimum shot length is $\ell_{min} = 1$ (no merging) everywhere. Everything else is as in the main study: existing best checkpoints (EMA weights selected by SHOT validation F1), sliding windows (T = 128, stride 64) averaged back to a per-video track, threshold grids $\tau_c \in \{0.10, 0.15, \ldots, 0.90\}$ and $\tau_g \in \{0.3, 0.4, 0.5, 0.6\}$, the decision rule (dual-label models: boundary if $p_y > \tau_c$, gradual if $p_z > \tau_g$, track = gradual OR cut; midpoint-only and unified models: τ_c only) and scoring with ±2-frame scene matching, with precision, recall and F1 pooled over all videos. Numbers are expected to be lower than with per-test-set tuning (Protocol A), especially on ClipShots and BBC. Single training seed (42) per model.

1. Validation data and threshold selection

The SHOT validation split (15 % of the 448 training videos, 67 videos) was previously available only as pre-cut training clips. For Protocol B these 67 videos were decoded completely, with the same decoder, resizing (48 × 64), window layout and window averaging as the SHOT test set (1,270 windows). The label file is the only ground truth available for these videos and is not the authors' final annotation, so validation F1 is noisier than test F1, and the thresholds it selects are noisier too. The validation videos were also used for early stopping and checkpoint selection, so validation F1 is slightly optimistic.

Validation-selected thresholds and validation F1 are listed in section 2.4.

2. Protocol B (thresholds selected once on the 67 SHOT validation videos, applied unchanged)

Each model's τ_c and τ_g were selected once, by maximising F1 on the 67 SHOT validation videos evaluated as full videos (same windows, averaging and scoring as the test sets, $\ell_{min} = 1$), and then applied unchanged to SHOT, ClipShots and BBC. Nothing was tuned on a test set. Validation F1 and the selected thresholds are in the thresholds table.

2.1 Supervision (Table 7a, seed 42)

ID	Description	SHOT F1	ClipShots F1	BBC F1
A0	Full, dual-label (primary)	0.8084	**0.7658**	0.9439
A1	Full, midpoint only	0.8313	0.7352	0.9597
A2	Full, unified single head	0.8350	0.7589	0.9473
A3	Reduced, midpoint only	0.8223	0.7562	**0.9713**
A4	Reduced, dual-label	**0.8364**	0.7627	0.9660

Delta	SHOT	ClipShots	BBC
A0 – A1 (full)	-0.0230	+0.0305	-0.0158
A4 – A3 (reduced)	+0.0141	+0.0065	-0.0054
A0 – A2 (dual vs unified)	-0.0267	+0.0069	-0.0034

2.2 Components, leave-one-out (Table 7b)

ID	Description	SHOT F1	ClipShots F1	BBC F1
A0	Full model	0.8084	0.7658	0.9439

ID	Description	SHOT F1	ClipShots F1	BBC F1
C1	w/o difference-based branch (B)	0.8357	0.7758	0.9444
C2a	w/o FrameSimilarity	**0.8408**	0.7265	0.9449
C2b	w/o ColorHistogramSimilarity	0.8260	0.7513	0.9546
C3	w/o temporal delta	0.8208	0.7709	0.9583
C4	w/o per-frame pathway (relations from Branch A)	0.8310	0.7920	0.9605
C5a	w/o pixel-level motion descriptor	0.8154	**0.7951**	0.9663
C5b	w/o edge-density head	0.8340	0.7638	**0.9680**
C6	w/o self-attention and gate	0.8147	0.7708	0.9534

Change in F1 relative to A0:

ID	ΔSHOT	ΔClipShots	ΔBBC
C1	+0.0274	+0.0101	+0.0005
C2a	+0.0325	-0.0393	+0.0009
C2b	+0.0176	-0.0144	+0.0107
C3	+0.0124	+0.0052	+0.0144
C4	+0.0226	+0.0262	+0.0165
C5a	+0.0070	+0.0294	+0.0224
C5b	+0.0257	-0.0020	+0.0241
C6	+0.0063	+0.0050	+0.0095

Largest single-component change on any dataset: 0.0393.

2.3 Build-up ladder

Step	Run	What is added	SHOT F1	ClipShots F1	BBC F1
0	M0_min	RGB encoder + MLP + dual head only	0.8124	0.7396	0.9502
1	M1_edge	+ edge-density head	0.8082	0.7633	0.9513
2	B_noatt	+ 4 fusion blocks, no attention	0.8259	0.7791	0.9525
3	A4	+ attention and gate (reduced model)	**0.8364**	0.7627	**0.9660**
4	S1_rel	+ FrameSimilarity, ColorHist, delta	0.8275	**0.7862**	0.9534
5	A0	+ diff branch, motion, per-frame pathway (full model)	0.8084	0.7658	0.9439

Step	SHOT	ClipShots	BBC
M0_min → M1_edge	-0.0042	+0.0236	+0.0011
M1_edge → B_noatt	+0.0177	+0.0159	+0.0012
B_noatt → A4	+0.0105	-0.0164	+0.0135
A4 → S1_rel	-0.0089	+0.0234	-0.0125
S1_rel → A0	-0.0191	-0.0204	-0.0095

2.4 Precision, recall and selected thresholds

ID	τ_c, τ_g (from validation)	Validation F1	SHOT P / R	ClipShots P / R	BBC P / R
A0	0.6, 0.4	0.7749	0.883 / 0.745	0.684 / 0.869	0.929 / 0.959
A1	0.55, –	0.7482	0.909 / 0.766	0.744 / 0.726	0.983 / 0.938
A2	0.55, –	0.7583	0.892 / 0.785	0.679 / 0.859	0.939 / 0.956
A3	0.5, –	0.7413	0.888 / 0.766	0.730 / 0.784	0.977 / 0.966
A4	0.45, 0.5	0.7405	0.865 / 0.810	0.665 / 0.894	0.948 / 0.985
C1	0.55, 0.3	0.7712	0.899 / 0.781	0.700 / 0.870	0.933 / 0.956

ID	τ_c, τ_g (from validation)	Validation F1	SHOT P / R	ClipShots P / R	BBC P / R
C2a	0.5, 0.4	0.7618	0.863 / 0.820	0.613 / 0.891	0.917 / 0.974
C2b	0.45, 0.3	0.7632	0.878 / 0.780	0.648 / 0.893	0.927 / 0.983
C3	0.55, 0.5	0.7588	0.863 / 0.783	0.685 / 0.882	0.950 / 0.966
C4	0.55, 0.5	0.7664	0.887 / 0.782	0.723 / 0.876	0.953 / 0.968
C5a	0.55, 0.6	0.7581	0.898 / 0.747	0.739 / 0.861	0.968 / 0.964
C5b	0.45, 0.5	0.7672	0.863 / 0.807	0.665 / 0.898	0.952 / 0.985
C6	0.55, 0.5	0.7539	0.862 / 0.772	0.686 / 0.880	0.946 / 0.961
M0_min	0.5, 0.6	0.7235	0.834 / 0.792	0.686 / 0.803	0.972 / 0.930
M1_edge	0.5, 0.6	0.7363	0.879 / 0.748	0.720 / 0.813	0.973 / 0.931
B_noatt	0.55, 0.5	0.7387	0.872 / 0.784	0.704 / 0.872	0.949 / 0.956
S1_rel	0.5, 0.4	0.7598	0.894 / 0.770	0.717 / 0.871	0.931 / 0.977

A1, A2, A3 and their seeds use τ_c only.

2.5 Table 8 — Per-type analysis on ClipShots

ID	Cut recall	Gradual recall	Gradual F1	Gradual IoU
A0	0.8712	0.8640	0.6488	0.7338
A1	0.8810	0.3968	n/a	n/a
A2	0.8718	0.8331	n/a	0.7218

2.6 Table 9 — Recall by gradual-transition duration on ClipShots

ID	≤ 10 frames	11–30 frames	> 30 frames
A0	0.8341 (n = 832)	0.8759 (n = 1241)	0.9079 (n = 228)
C6	0.8257 (n = 832)	0.8767 (n = 1241)	0.8421 (n = 228)

C6 – A0: ≤ 10 -0.0084, 11–30 +0.0008, > 30 -0.0658.

3. Protocol B compared with Protocol A (ℓ_min = 1)

Protocol A selects thresholds on each test set (optimistic); Protocol B does not. F1 under each protocol and the difference:

ID	SHOT A	SHOT B	Δ	ClipShots A	ClipShots B	Δ	BBC A	BBC B	Δ
A0	0.8449	0.8084	-0.0366	0.8001	0.7658	-0.0344	0.9719	0.9439	-0.0279
A1	0.8478	0.8313	-0.0165	0.7609	0.7352	-0.0257	0.9698	0.9597	-0.0101
A2	0.8509	0.8350	-0.0159	0.7709	0.7589	-0.0120	0.9473	0.9473	-0.0000
A3	0.8312	0.8223	-0.0089	0.7606	0.7562	-0.0044	0.9714	0.9713	-0.0000
A4	0.8370	0.8364	-0.0006	0.8045	0.7627	-0.0418	0.9721	0.9660	-0.0061
C1	0.8536	0.8357	-0.0179	0.7942	0.7758	-0.0184	0.9701	0.9444	-0.0257
C2a	0.8457	0.8408	-0.0048	0.8009	0.7265	-0.0744	0.9681	0.9449	-0.0233
C2b	0.8361	0.8260	-0.0101	0.8077	0.7513	-0.0564	0.9730	0.9546	-0.0184
C3	0.8363	0.8208	-0.0155	0.7956	0.7709	-0.0247	0.9728	0.9583	-0.0144
C4	0.8420	0.8310	-0.0110	0.8068	0.7920	-0.0148	0.9699	0.9605	-0.0094
C5a	0.8487	0.8154	-0.0333	0.8021	0.7951	-0.0070	0.9715	0.9663	-0.0053
C5b	0.8454	0.8340	-0.0113	0.7967	0.7638	-0.0329	0.9733	0.9680	-0.0053
C6	0.8364	0.8147	-0.0217	0.7969	0.7708	-0.0262	0.9667	0.9534	-0.0133
M0_min	0.8169	0.8124	-0.0045	0.7555	0.7396	-0.0159	0.9543	0.9502	-0.0041

ID	SHOT A	SHOT B	Δ	ClipShots A	ClipShots B	Δ	BBC A	BBC B	Δ
M1_edge	0.8261	0.8082	-0.0179	0.7672	0.7633	-0.0039	0.9584	0.9513	-0.0071
B_noatt	0.8351	0.8259	-0.0093	0.7984	0.7791	-0.0193	0.9682	0.9525	-0.0157
S1_rel	0.8392	0.8275	-0.0117	0.8054	0.7862	-0.0192	0.9706	0.9534	-0.0171

Mean change over the 17 models: SHOT -0.0146 (range -0.0366 to -0.0006), ClipShots -0.0254 (range -0.0744 to -0.0039), BBC -0.0120 (range -0.0279 to -0.0000). No model scores higher under Protocol B than under Protocol A on any test set, as expected, since Protocol A picks the best thresholds on the test set itself.

4. Interpretation check (0.01 band)

Statement	Value	Holds?
A0 – A1 on ClipShots ≥ 0.01	+0.0305	**yes**
A4 – A3 on ClipShots ≥ 0.01	+0.0065	**no**
A0 – A2 on ClipShots ≥ 0.01	+0.0069	**no**
A0 – A2 on BBC ≥ 0.01	-0.0034	**no**
No leave-one-out change (C1–C6) reaches 0.01 on any test set	largest 0.0393	**no**

Differences below 0.01 F1 are treated as inconclusive. Under Protocol B the thresholds themselves add noise: a model's validation-selected τ_c / τ_g can sit far from its best test thresholds, so single-model differences (for example in the leave-one-out table) are less reliable than under Protocol A.

5. SHOT validation F1 per epoch (training histories)

Epoch	A1	A0
1	0.7029	0.6988
2	0.7499	0.7514
3	0.7535	0.7543
4	0.7628	0.7774
5	0.7807	0.7758
6	0.7796	0.7722
7	**0.7863**	0.7725
8	0.7785	0.7748
9	0.7695	0.7654
10	0.7852	0.7838
11	0.7843	0.7793
12	0.7783	0.7843
13	0.7824	0.7821
14	0.7704	0.7681
15	0.7840	0.7885
16	0.7735	0.7926
17	0.7846	0.7810
18		0.7932
19		0.7946
20		0.7960
21		**0.8029**
22		0.7935
23		0.7907
24		0.7953

Epoch	A1	A0
25		0.7854
26		0.7847
27		0.7913
28		0.7951
29		0.7810
30		0.7886
31		0.7870

Best epoch in bold. This is the validation F1 used for early stopping (pre-cut validation clips, fixed thresholds), not the full-video validation F1 of section 1.